

REFINED QUALITY ASSURANCE TESTING DOCUMENTATION (QATD)

UMHackathon 2026

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Document Control

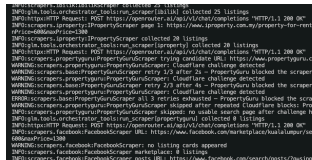
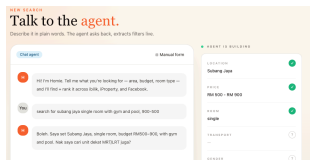
Field	Detail
System Under Test (SUT)	UMHackathon 2026 - Team Big Guys
Team Repo URL	https://github.com/FlyingShuriken/homie
Live Deployment URL	http://54.188.162.45/

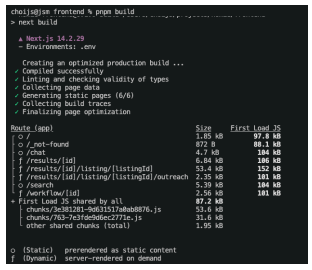
Objective:

The primary objective of this testing documentation is to verify that **Homie** can execute its core rental-search workflow reliably: accept search filters, run multi-source listing collection, normalize and score listings, present ranked results, and support inquiry drafting and handoff. A secondary objective is to identify release blockers and operational risks before hackathon judging, especially around scraper stability, AI fallback behavior, and frontend build readiness.

FINAL ROUND (Execution, RCA & Hard Evidence)

1. Test Execution & Evidence (Manual)

Test Case ID	Test Type & Mapped Feature	Test Description	Expected Result	Actual Result	Status	Proof of Execution (Required for P0/Critical tests)
TC-01	Happy Case (Core Workflow): filters -> seeded multi-source collection -> normalize -> score -> report	Verify that Homie can process a Cheras rental search through the real backend tool implementations using fixture mode for all four sources.	Filters are validated; ibilik, iProperty, PropertyGuru, and Facebook fixtures load; listings are normalized and scored; report is generated; workflow can finish without live GLM/network dependencies.	Fixture harness loaded 55 raw listings and 40 normalized listings from all four sources. Score stage processed 40 listings with top_score 100.0/100.	Pass	
TC-02	Specific Negative Case: invalid price range validation	Verify that the search API/form validation autocorrect impossible budgets where price_max is less than or equal to price_min.	The GLM API auto correct the wrongly input value to the correct form	GLM auto correct RM 900 - 500 to RM 500 - 900	Pass	

Test Case ID	Test Type & Mapped Feature	Test Description	Expected Result	Actual Result	Status	Proof of Execution (Required for P0/Critical tests)
TC-03	NFR / Release Readiness: frontend production build	Verify that the Next.js frontend can be built for deployment and that all dynamic result/workflow routes compile.	pnpm build completes successfully with no TypeScript or production-build blocker.	pnpm build completed successfully. Routes generated: /, /chat, /search, /workflow/[id], /results/[id], /results/[id]/listing/[listingId], and /outreach.	Pass	 <pre> chaitojin:frontend % pnpm build > next build A next.js 14.2.20 Environment: .env Creating an optimized production build ... Compiled successfully Linting and checking validity of types Collecting page data Generating static pages (6/6) Collecting build traces Finalizing page optimization Route (app) Size First Load JS ┌ ○ / 1.25 kB 97.8 kB ├ ○ /_not-found 872 B 80.1 kB ├ ○ /chat 4.7 kB 184 kB ├ /results/[id] 8.84 kB 185 kB ├ /results/[id]/listing/[listingId] 53.4 kB 152 kB ├ /results/[id]/listing/[listingId]/outreach 7.35 kB 183 kB ├ ○ /search 5.35 kB 184 kB ├ /workflow/[id] 2.56 kB 181 kB └ First Load JS shared by all chunks/6283251-5d63537f0a080876.js 23.6 kB chunks/763-7d3f6d8dec2771e.js 21.6 kB other shared chunks (total) 1.55 kB ○ (Static) prerendered as static content └ (Dynamic) server-rendered on demand </pre>
TC- AI-01	AI Output / Fallback Validation: GLM unavailable during report/explanation enrichment	Verify that GLM-dependent explanation/report enrichment fails safely when GLM_API_KEY is not set in the local QA environment.	System keeps deterministic scoring results and uses fallback explanations/report instead of crashing.	Fixture harness logged GLM enrich/report failures, then kept fallback explanations and produced a fallback report successfully.	Pass	

2. Automated Testing Pipeline

2.1. Unit Testing (Core Logic & Functions)

Unit testing isolates backend functions and route-level helpers so core behavior can be verified without live scraper pages, GLM network calls, Telegram MTProto, or a production database. External APIs are mocked or replaced with temporary SQLite/test fixtures where appropriate.

- **Unit Test**
 - Current Scope: Individual unit (function/method/class) behavior in isolation.
 - Execution: Automated and fast; run frequently (locally and in CI) with minimal setup.
 - Isolation: High—external dependencies (DB, filesystem, network, time, randomness, other services) are mocked/stubbed/faked.
 - Pass Condition: All assertions meet expected outcomes; test is deterministic and repeatable (same inputs $\Rightarrow$ same results).

Unit Test ID	Function/Component Tested	Test Scenario	Expected Outcome	Actual Result	Status
UT-01	extract_gender_restriction() / apply_deterministic_extraction()	Detect male/female/mixed tenant restrictions across English and Bahasa Malaysia listing text while avoiding false positives such as “female landlord”.	Correct normalized gender value or None; pre-parsed explicit values are preserved.	Parameterized tests and fixture examples passed.	Pass
UT-02	Google Places enrichment helpers	Map mocked Places Search and Place Details responses into Homie listing google_place fields and reject low-confidence matches.	Correct place_id, rating, review serialization, and low-confidence rejection.	test_google_places.py passed all cases.	Pass
UT-03	SSE replay buffer	A late client connects after the workflow queue has been removed but replay events remain buffered.	stream_progress returns buffered events instead of 404.	test_stream_progress_replays_completed_session_without_queue passed.	Pass

UT-04	cleanup_expired_records()	Expired sessions/listings should remove related outreach events and Telegram conversations without deleting active records.	Expired rows removed; active rows preserved.	SQLite-isolated hardening test passed.	Pass
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Execution Proof:

```

choijs@jsm frontend % pytest
===== test session starts =====
platform darwin -- Python 3.13.5, pytest-8.4.0, pluggy-1.6.0
rootdir: /Users/choijs/projects/homie/frontend
plugins: mock-3.15.1, anyio-4.9.0, asyncio-1.3.0
asyncio: mode=Mode.STRICT, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scope=function
collected 0 items

===== no tests ran in 0.01s =====
choijs@jsm frontend % cd ..
choijs@jsm homie % cd backend
choijs@jsm backend % pytest
===== test session starts =====
platform darwin -- Python 3.13.5, pytest-8.4.0, pluggy-1.6.0
rootdir: /Users/choijs/projects/homie/backend
configfile: pyproject.toml
plugins: mock-3.15.1, anyio-4.9.0, asyncio-1.3.0
asyncio: mode=Mode.STRICT, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scope=function
collected 33 items

tests/test_gender_extraction.py ..... [ 72%]
tests/test_google_places.py ... [ 81%]
tests/test_hardening.py ..... [100%]

===== 33 passed in 0.95s =====

```

2.2. Integration & API Testing (System Modules)

- **Integration Test**
 - **Scope:** Search session creation, orchestrator-driven tool execution, result persistence, outreach draft generation, and handoff fallback
 - **Execution:** Controlled backend integration smoke tests using FastAPI TestClient
 - **Workflow:** Temporary SQLite database, DEMO_SEED=true, deterministic mocked AI responses, and repository fixture listings
 - **Pass Condition:** Pipeline reaches complete, returns listings, produces summary text, and allows inquiry draft generation without crashing

Test ID	Integration / Point Tested	Test Scenario	Expected Outcome	Actual Result	Status
IT-01	Scraper fixtures -> normalize_listings -> score_listings -> generate_report	Run backend tool pipeline in DEMO_SEED fixture mode for Cheras single-room search.	All four fixture sources load; normalized listings are scored; fallback report generated.	Loaded 7 iBilik, 7 iProperty, 3 PropertyGuru, and 7 Facebook listings. Normalized/scored 24 listings; avg_score 59.4; top_score 81.0.	PASS
IT-02	FastAPI result serializer -> frontend result contract	Serialize a DBListing-like row containing Google Places fields.	Frontend receives google_place object with place_id, rating, and reviews in expected shape.	serialize_listing_response returned expected google_place payload in test_google_places.py.	PASS
IT-03	Backend SSE endpoint -> workflow UI consumption model	Replay completed workflow events when the queue is gone but buffered session events exist.	Late-joining frontend EventSource can still display events.	SSE replay unit/integration hardening test passed.	PASS

Execution Proof:

```

===== test session starts =====
platform darwin -- Python 3.10.20, pytest-9.0.3, pluggy-1.6.0
rootdir: /Users/choijs/projects/homie/backend
configfile: pyproject.toml
plugins: asyncio-1.3.0, anyio-4.13.0
asyncio: mode=strict, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_sc
ope=function
collected 3 items

tests/test_google_places.py ... [100%]

===== 3 passed in 0.40s =====

```

```

~/projects/homie/backend git:(master) 4 • +112 -5 (0.71s)
uv run pytest tests/test_hardening.py::test_stream_progress_replays_completed_session_without_queue
===== test session starts =====
platform darwin -- Python 3.10.20, pytest-9.0.3, pluggy-1.6.0
rootdir: /Users/choijs/projects/homie/backend
configfile: pyproject.toml
plugins: asyncio-1.3.0, anyio-4.13.0
asyncio: mode=strict, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_sc
ope=function
collected 1 item

tests/test_hardening.py . [100%]

===== 1 passed in 0.40s =====

```

```

uv run pytest tests/test_hardening.py::test_telegram_demo_outreach_respects_session_opt_out
===== test session starts =====
platform darwin -- Python 3.10.20, pytest-9.0.3, pluggy-1.6.0
rootdir: /Users/choijs/projects/homie/backend
configfile: pyproject.toml
plugins: asyncio-1.3.0, anyio-4.13.0
asyncio: mode=strict, debug=False, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_sc
ope=function
collected 1 item

tests/test_hardening.py . [100%]

===== 1 passed in 0.42s =====

```


3. Defect Log & Fix Traceability (Root Cause Analysis)

Log the bugs you **fixed**. Include the exact Git Commit Hash to prove the fix. AI cannot fake your codebase's commit history.

Bug ID	Bug Description	Steps to Reproduce	Console/Error Log Snippet	Root Cause Analysis (Why did it break?)	Fix Commit Hash / PR Link
DEF-01	Transit station mentions were not normalized consistently, causing weak transport scoring and unclear transit display.	Run normalization on listings mentioning MRT/LRT in free text or Bahasa Malaysia; inspect nearby_transport output.	Before fix: station names could be missing, generic, or inconsistent across scrapers.	Extractor lacked a complete Malaysian station reference and deterministic station pre-pass; GLM had too little context to match local station names reliably.	https://github.com/FlyingShuriken/homie/commit/05131c680bd3eccd4eca0b073dcc4ba245e394a1 https://github.com/FlyingShuriken/homie/commit/a807962eb62af4da5ac32c6f6d22795f58bba922
DEF-02	Listing detail page lacked map/transit verification context for transport claims.	Open a listing with lat/lng or nearby transit data and inspect detail page.	Before fix: no visual map pins or walking-time verification fields available.	Backend stored listing/transit data but did not enrich transit stops or render them in the frontend detail view.	https://github.com/FlyingShuriken/homie/commit/60aebae35cd5bab840769f50e90d48138d08a58bbbc9d8003d95bb5 https://github.com/FlyingShuriken/homie/commit/e2d02341d720fc488acf79b48bbc9d8003d95bb5
DEF-03	Score explanations were too generic; users could not see why each score dimension was awarded.	Run scoring on fixture listings and open score breakdown in listing detail page.	Before fix: explanation text summarized the total but dimension-level comments were missing.	The scoring response persisted numeric breakdown but not concise per-dimension comments for frontend display.	https://github.com/FlyingShuriken/homie/commit/b336f5369510ca3abcebfa487563b7841dcbbc15

4. Known Issues & Deferred Technical Debt

Bug ID	Description & Impact (Bug or Technical Debt)	Reason for Deferral	GitHub Issue Link
KTD-01	Orchestrator can mark a session complete if the model returns plain text instead of calling the finish tool. Impact: possible empty "complete" result if GLM tool calling fails unexpectedly.	Requires deterministic required-step guard or non-LLM fallback pipeline; deferred because current demo path assumes GLM tool-calling model is configured.	https://github.com/FlyingShuriken/homie/issues/9
KTD-02	Parking and pet-friendly filters are collected in the UI/API but are not currently included in backend scoring. Impact: users may think these preferences affect ranking when they only appear as listing fields.	Core scoring focused on price, location, room type, furnishing, contact, images, gender, and transport first.	https://github.com/FlyingShuriken/homie/issues/7
KTD-03	Google geocoding/walking verification helpers exist, but the active orchestrator normalization path does not fully execute listing/station geocoding. Impact: actual_walk_minutes may be empty and transport scoring falls back to partial credit.	Google API key/quota dependency and time constraints; kept Places enrichment optional for demo.	https://github.com/FlyingShuriken/homie/issues/8

5. Live Demo / UAT Risks

Homie is reliable when configured with the expected environment variables and/or demo-seed mode, but live external services can still introduce variability.

The demo should be run with the PM2 deployment path, a configured database, GLM credentials, and pre-authenticated Telegram session if demo outreach is shown.

- Risk 1: Live scraping may be blocked or return partial data, especially iProperty, PropertyGuru, and Facebook due to Cloudflare, login requirements, or DOM changes. For judging, keep DEMO_SEED=true ready as the stable fallback path; treat live scraping as best-effort.
- Risk 2: GLM_API_KEY, GLM_BASE_URL, and DATABASE_URL must be configured before starting the backend. Without GLM credentials, the top-level orchestrator cannot drive a live search, although deterministic fallback logic exists inside normalization/scoring/report tools.
- Risk 3: Telegram outreach is demo-target based only. Messages are routed to TELEGRAM_DEMO_TARGET through an authenticated Telethon session; Homie does not directly message real landlords during judging.
- Risk 4: The backend is a single-process PM2 demo service with in-memory SSE queues. Do not restart the backend while a search is running; refresh completed results through the results page if needed.
- Risk 5: Google Maps ratings, reviews, map pins, and walking-time estimates require GOOGLE_MAPS_API_KEY and NEXT_PUBLIC_GOOGLE_MAPS_API_KEY. If those keys are absent or quota-limited, the listing still works but enrichment panels may be hidden or partial.